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Course: Digital Electronics	Instructor: Nuzhath Farhana
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Design and Implementation of BCD to Excess-3 Code Converter using Karnaugh Map Minimization and SimulIDE Software

Objective:

To design a BCD (Binary-Coded Decimal) to Excess-3 code converter circuit using Karnaugh Map simplification and simulate the logic circuit in SimulIDE.

Software Used:

- SimulIDE v1.x
- Platform: Windows

Logic:

Let the BCD input be A B C D (A is MSB, D is LSB).

We can generate Excess-3 code W X Y Z by adding binary 3 (0011) to the BCD input using a 4-bit binary adder.

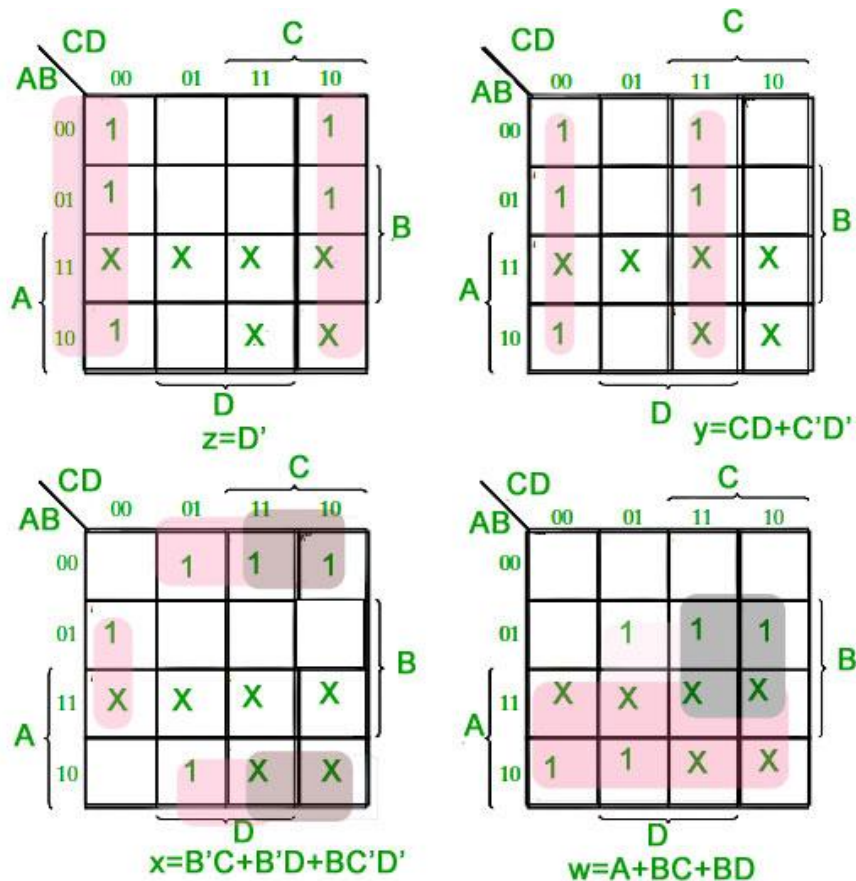
Procedure:

1. Launch SimulIDE software.
2. From the component library, drag and drop:
 - 4 Toggle Switches for inputs A, B, C, D.
 - 4 LEDs for displaying output W, X, Y, Z.
 - Logic gates to build 4-bit binary adder.
3. Connect the BCD inputs to the adder.
4. Add binary 3 (0011) to the input using the adder.
5. Connect the output of the adder to LEDs.
6. Provide Vcc and Ground to the components.
7. Run simulation and test for all BCD inputs (0000 to 1001).

Truth Table:

Decimal	BCD (A B C D)	Excess-3 (W X Y Z)
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

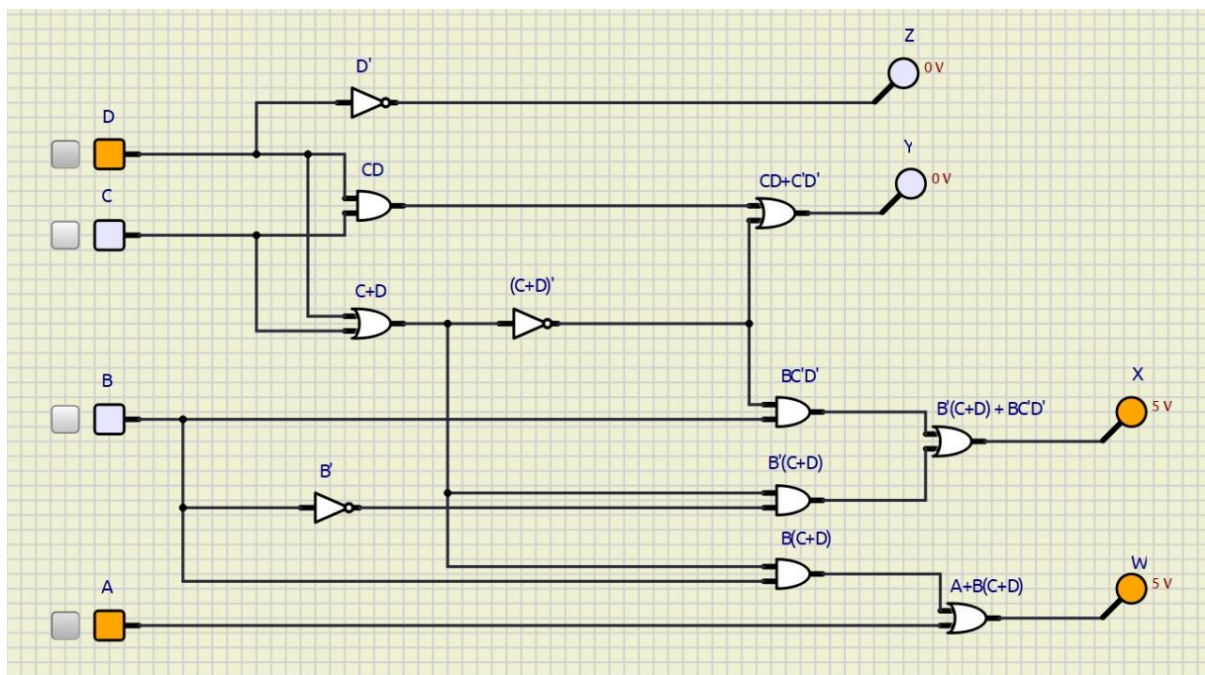
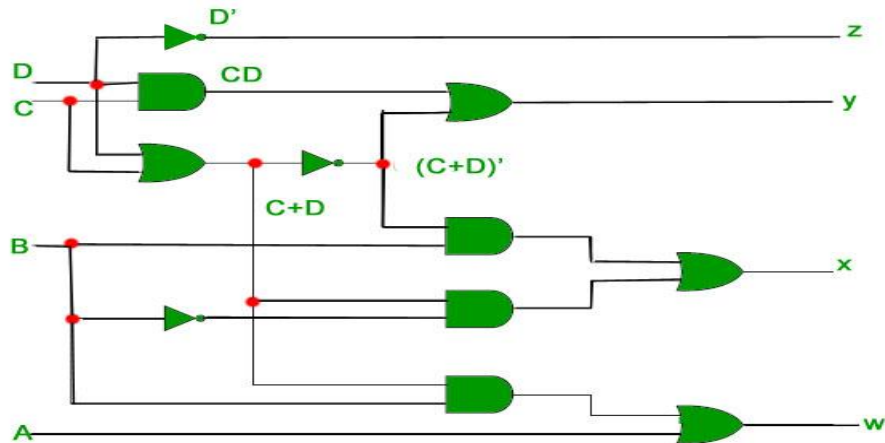
To find the corresponding digital circuit, we will use the K-Map technique for each of the Excess-3 code bits as output with all of the bits of the BCD number as input.



Corresponding minimized Boolean expressions for Excess-3 code bits:

$$w = A + BC + BD$$

$$x = B'C + B'D + BC'D'$$



Result:

The BCD to Excess-3 code converter was successfully designed and simulated using SimulIDE. The output matched the expected Excess-3 code for all valid inputs.

Design and Implementation of Binary to Gray Code Converter using Karnaugh Map and SimulIDE

Objective:

To design a Binary to Gray Code Converter circuit using Karnaugh Map minimization, and simulate it using SimulIDE software.

Software Used:

- SimulIDE v1.x
- Platform: Windows

Let b_0, b_1, b_2 and b_3 be the bits representing the binary numbers, where b_0 is the LSB and b_3 is the MSB, and Let g_0, g_1, g_2 , and g_3 be the bits representing the gray code of the binary numbers, where g_0 is the LSB and g_3 is the MSB.

The truth table for the conversion is

Binary (B3 B2 B1 B0)	Gray Code (G3 G2 G1 G0)
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

To find the corresponding digital circuit, we will use the K-Map technique for each of the gray code bits as output with all of the binary bits as input.

K-map for g0:

		b1,b0			
		00	01	11	10
b3,b2	00	0	1	0	1
	01	0	1	0	1
	11	0	1	0	1
	10	0	1	0	1

K-map for g1:

		b1,b0			
		00	01	11	10
b3,b2	00	0	0	1	1
	01	1	1	0	0
	11	1	1	0	0
	10	0	0	1	1

K-map for g2:

		b1,b0			
		00	01	11	10
b3,b2	00	0	0	0	0
	01	1	1	1	1
	11	0	0	0	0
	10	1	1	1	1

K-map for g3:

		b1,b0			
		00	01	11	10
b3,b2	00	0	0	0	0
	01	0	0	0	0
	11	1	1	1	1
	10	1	1	1	1

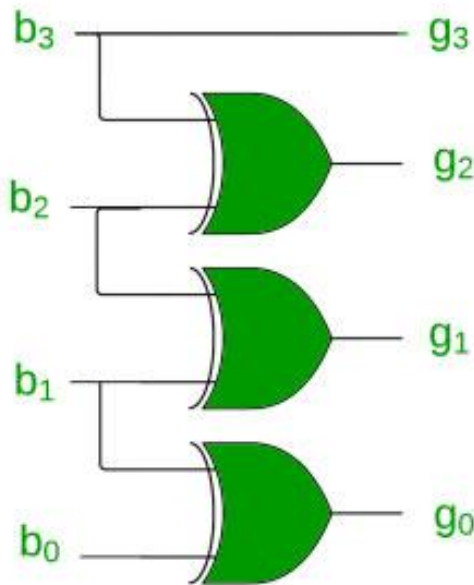
Corresponding Minimized Boolean Expressions and Digital Circuit for Gray Code Bits

$$g_0 = b_0b_1' + b_1b_0' = b_0 \oplus b_1$$

$$g_1 = b_2b_1' + b_1b_2' = b_1 \oplus b_2$$

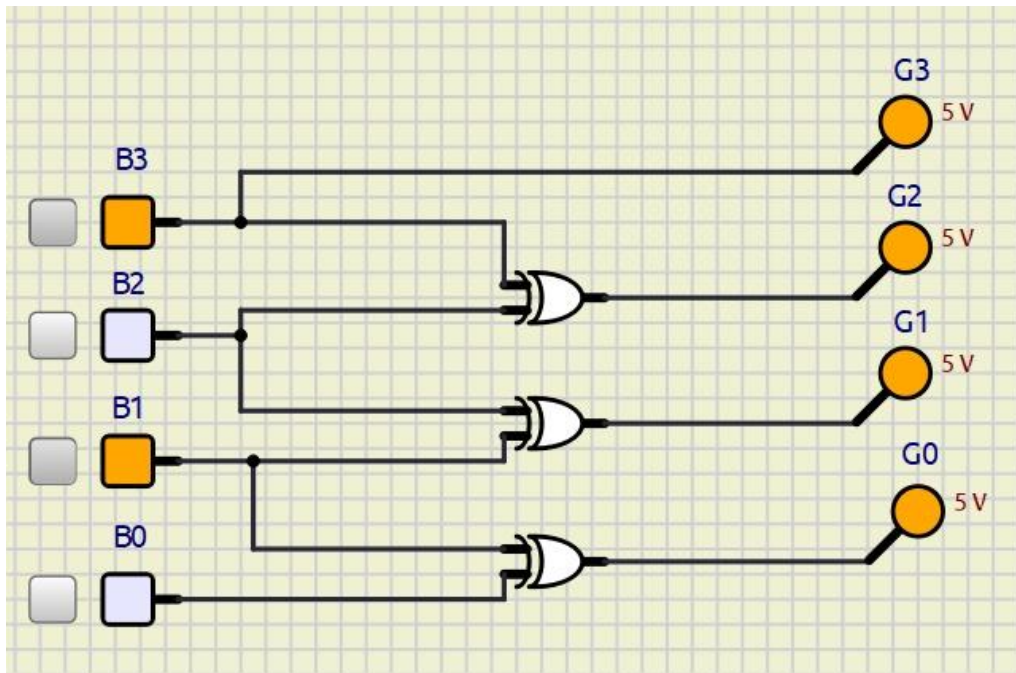
$$g_2 = b_2b_3' + b_3b_2' = b_2 \oplus b_3$$

$$g_3 = b_3$$



Procedure in SimulIDE:

1. Open SimulIDE software.
2. Drag and drop:
 - 4 Toggle Switches: for binary inputs A, B, C, D
 - 4 LEDs: for Gray code outputs W, X, Y, Z
 - Logic Gates: XOR gates (for X, Y, Z), direct wire for W
3. Build the logic:
 - $W = A$
 - $X = A \text{ XOR } B$
 - $Y = B \text{ XOR } C$
 - $Z = C \text{ XOR } D$
4. Connect the inputs to gates, and outputs to LEDs.
5. Power the circuit with Vcc and GND.
6. Run the simulation and test for all 16 combinations.



Result:

The Binary to Gray Code Converter was successfully designed using minimized Boolean expressions via K-Map and implemented in SimulIDE. The output Gray code matched the expected values for all binary inputs.

Design and Implementation of Gray Code to Binary Code Converter using Karnaugh Map and SimulIDE Software

Objective:

To design a Gray to Binary Code Converter using Karnaugh Map simplification and implement it using SimulIDE software.

Software Used:

- SimulIDE v1.x
- Platform: Windows

Converting gray code back to binary can be done in a similar manner. Let b_0, b_1, b_2 and b_3 be the bits representing the binary numbers, where b_0 is the LSB and b_3 is the MSB, and Let g_0, g_1, g_2 , and g_3 be the bits representing the gray code of the binary numbers, where g_0 is the LSB and g_3 is the MSB.

Truth table:

Gray Code (G3 G2 G1 G0)	Binary (B3 B2 B1 B0)
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0000	0000
0001	0001
0011	0010
0010	0011
0110	0100
0111	0101
0101	0110
0100	0111
1100	1000
1101	1001
1111	1010
1110	1011
1010	1100
1011	1101
1001	1110
1000	1111

Using K-map to get back the binary bits from the gray code.

K-map for b0:

		g ¹ ,g ⁰			
		00	01	11	10
g ³ ,g ²	00	0	1	0	1
	01	1	0	1	0
	11	0	1	0	1
	10	1	0	1	0

K-map for b1:

		g ¹ ,g ⁰			
		00	01	11	10
g ³ ,g ²	00	0	0	1	1
	01	1	1	0	0
	11	0	0	1	1
	10	1	1	0	0

K-map for b2:

		g ¹ ,g ⁰			
		00	01	11	10
g ³ ,g ²	00	0	0	0	0
	01	1	1	1	1
	11	0	0	0	0
	10	1	1	1	1

K-map for b3:

		g ¹ ,g ⁰			
		00	01	11	10
g ³ ,g ²	00	0	0	0	0
	01	0	0	0	0
	11	1	1	1	1
	10	1	1	1	1

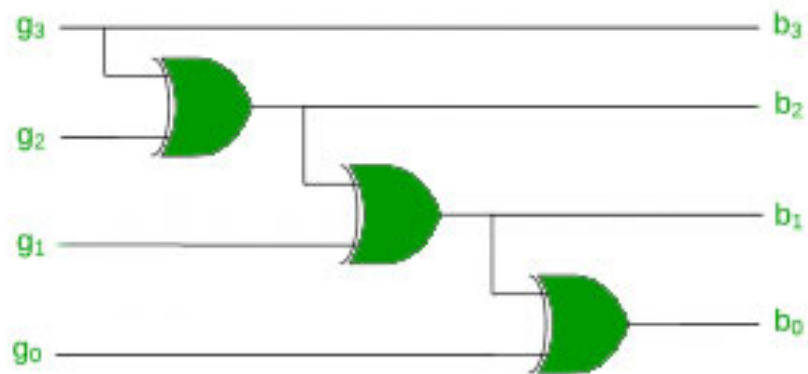
Corresponding Minimized Boolean Expressions

$$b_3 = g_3$$

$$b_2 = g_3 \oplus g_2$$

$$b_1 = g_2 \oplus g_1$$

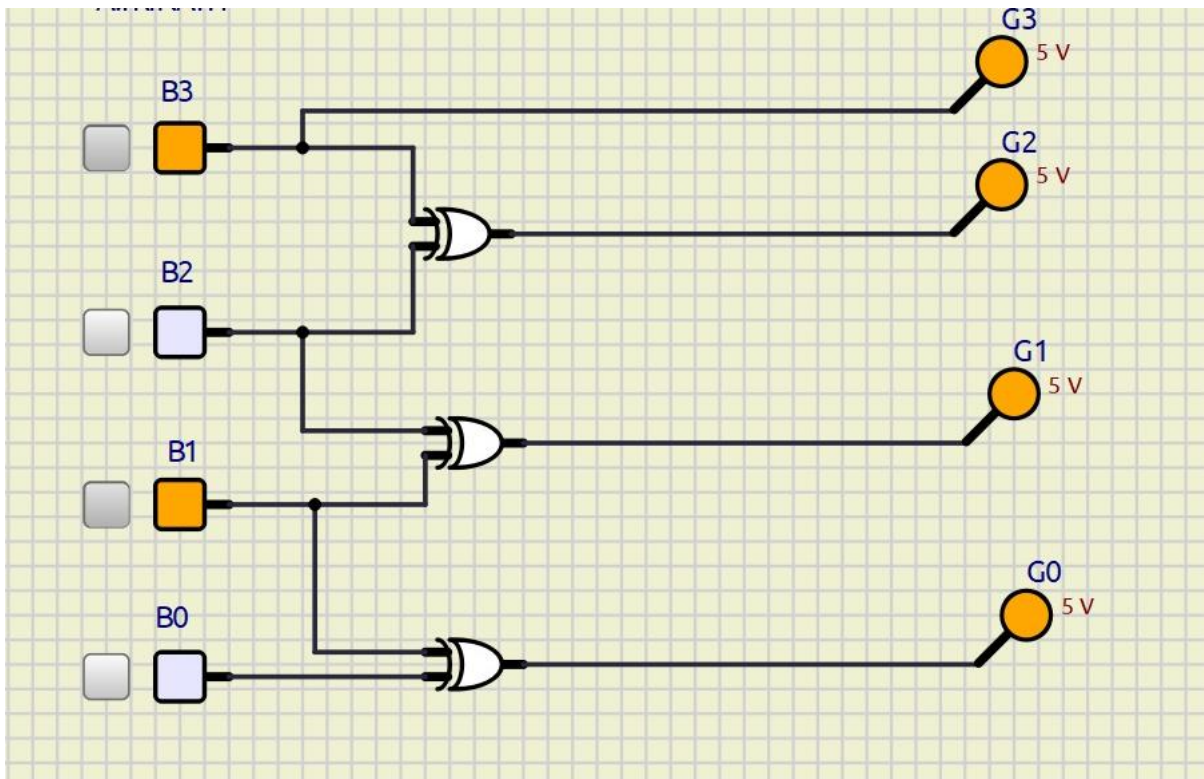
$$b_0 = g_1 \oplus g_0$$



Procedure in SimulIDE:

1. Launch SimulIDE.
2. Drag and place:
 - 4 Toggle switches for Gray inputs (W, X, Y, Z)
 - 4 LEDs for Binary outputs (A, B, C, D)
 - Logic gates: XOR gates
3. Build logic circuits as per:

- $A = W$
 - $B = W \oplus X$
 - $C = W \oplus X \oplus Y$
 - $D = W \oplus X \oplus Y \oplus Z$
4. Connect Gray inputs to gates, and outputs to LEDs.
 5. Connect power supply and ground.
 6. Run simulation and verify outputs for all 16 Gray code values.

**Result:**

The Gray to Binary Code Converter was successfully designed using Karnaugh Map simplified Boolean expressions and implemented in SimulIDE. The circuit worked as expected for all valid Gray code inputs.